

LAB EXPERIMENT RECORD

Set 3: Employee Performance Evaluation

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To evaluate employee performance data using a line chart, bar chart, and scatter plot in R, and to build an interactive dashboard combining the line chart and bar chart using R Shiny.

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

PROGRAM 1: Line Chart of Performance Trend

Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

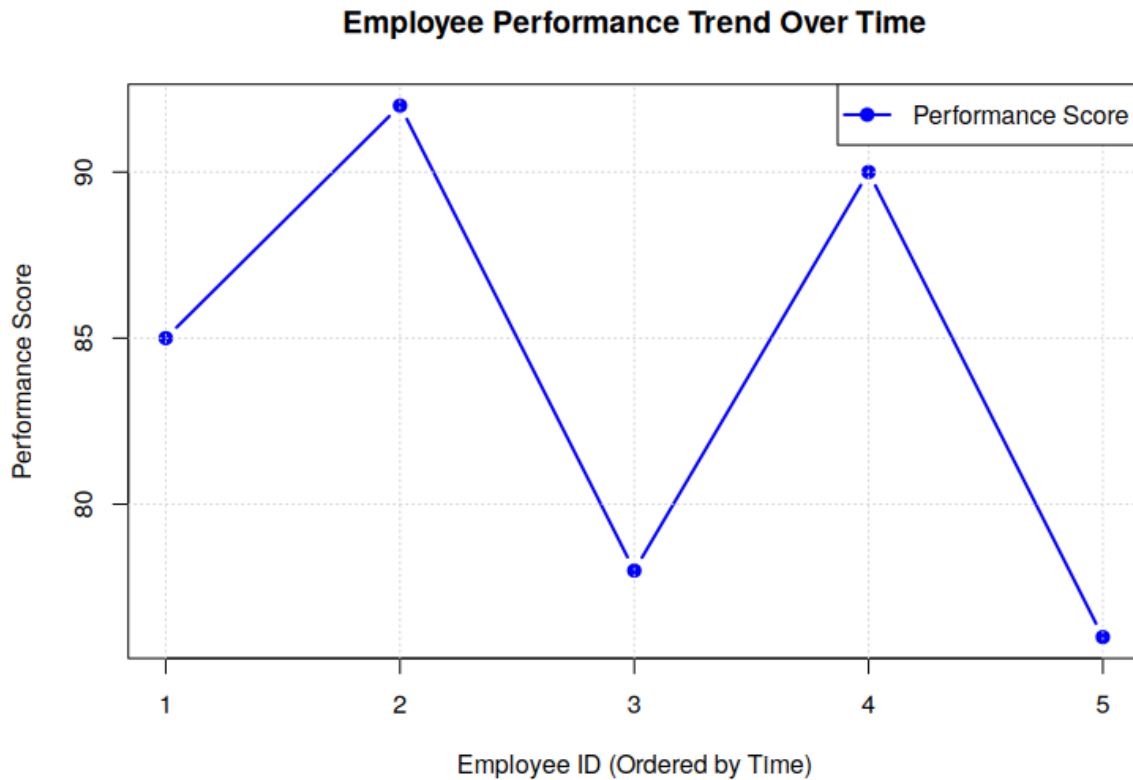
```
# Create dataset for Set 3
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = factor(c("Sales", "HR", "Marketing", "Sales", "HR")),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

print(employee_data)

# 1. Line Chart for Performance Trend
plot(employee_data$Employee_ID, employee_data$Performance_Score,
  type = "b",
  col = "blue",
  lwd = 2,
  pch = 19,
  xlab = "Employee ID (Ordered by Time)",
  ylab = "Performance Score",
  main = "Employee Performance Trend Over Time")
```

```
grid()

# Add Legend
legend("topright", legend = c("Performance Score"),
      col = c("blue"), lty = 1, pch = 19, lwd = 2)
```



Output 1: Employee Performance Trend Over Time

Observation: The performance scores move up and down across the employees. The highest score is 92 (Employee 2) and the lowest is 76 (Employee 5), so there is no steady increasing or decreasing trend.

PROGRAM 2: Bar Chart of Employees by Department

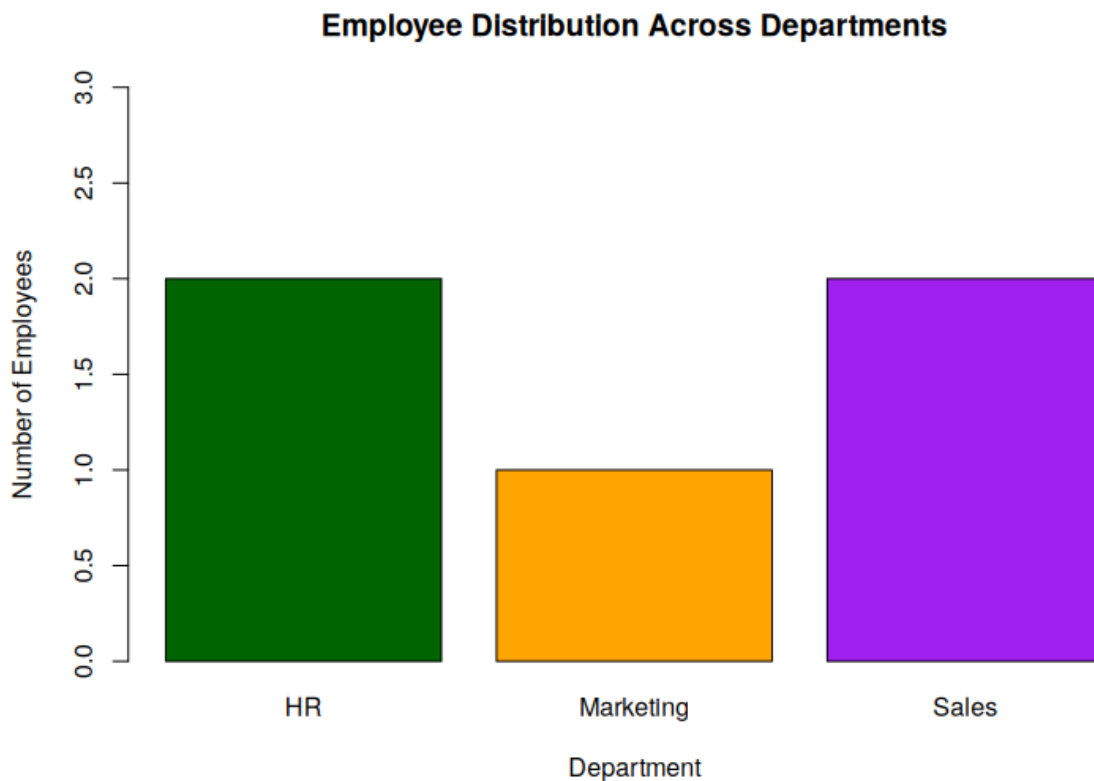
Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
# Create dataset for Set 3
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = factor(c("Sales", "HR", "Marketing", "Sales", "HR")),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

print(employee_data)

# Count employees per department
dept_counts <- table(employee_data$Department)

# 2. Bar Chart for Department Distribution
barplot(dept_counts,
  col = c("darkgreen", "orange", "purple"),
  xlab = "Department",
  ylab = "Number of Employees",
  main = "Employee Distribution Across Departments",
  ylim = c(0, 3))
```



Output 2: Employee Distribution Across Departments

Observation: HR and Sales each have two employees, while Marketing has only one employee.

PROGRAM 3: Scatter Plot of Years of Service vs Performance

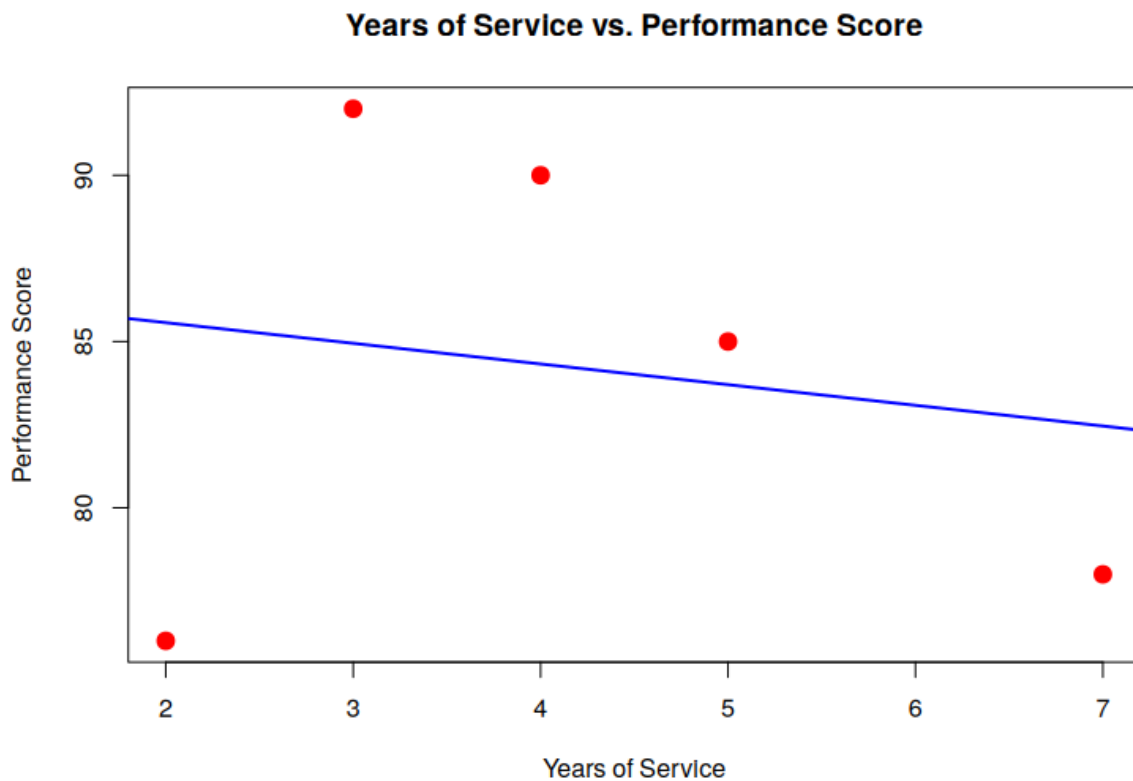
Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

```
# Create dataset for Set 3
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = factor(c("Sales", "HR", "Marketing", "Sales", "HR")),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

print(employee_data)

# 3. Scatter Plot for Years of Service vs Performance Score
plot(employee_data$Years_of_Service, employee_data$Performance_Score,
      col = "red",
      pch = 19,
      cex = 1.5,
      xlab = "Years of Service",
      ylab = "Performance Score",
      main = "Years of Service vs. Performance Score")

# Add a trend line
abline(lm(Performance_Score ~ Years_of_Service, data = employee_data), col = "blue", lwd = 2)
```



Output 3: Years of Service vs Performance Score

Insight: The trend line slopes slightly downward and the correlation coefficient is about -0.17 , which is a very weak negative relationship. Years of service do not clearly decide the performance score: the employee with 3 years scored the highest (92), while the employee with 7 years scored 78. With only five employees, this result is not conclusive.

PROGRAM 4: Interactive Dashboard for Employee Performance

Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 3
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("dept", "Department:",
        choices = c("HR", "Marketing", "Sales"),
        selected = c("HR", "Marketing", "Sales"))
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("linePlot", height = "340px")),
        column(6, plotOutput("barPlot", height = "340px"))
      )
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({ subset(employee_data, Department %in% input$dept) })

  output$linePlot <- renderPlot({
    d <- filtered()
    plot(d$Employee_ID, d$Performance_Score, type = "b", col = "blue", lwd = 2, pch = 19,
      xlim = c(1, 5), ylim = c(70, 95), xlab = "Employee ID (Ordered by Time)",
      ylab = "Performance Score", main = "Employee Performance Trend")
    legend("topright", legend = "Performance Score", col = "blue", lty = 1, pch = 19, bty = "n")
    grid()
  }, width = 420, height = 340, res = 96)

  output$barPlot <- renderPlot({
    counts <- table(factor(filtered()$Department, levels = c("HR", "Marketing", "Sales")))
    barplot(counts, col = c("darkgreen", "orange", "purple"), ylim = c(0, 3),
      xlab = "Department", ylab = "Number of Employees",
      main = "Employees by Department")
  }, width = 420, height = 340, res = 96)
}

shinyApp(ui = ui, server = server)
```

Employee Performance Dashboard



Output 4: Employee Performance Interactive Dashboard (R Shiny)

Observation: The dashboard shows the employee performance trend and the number of employees in each department side by side. Unticking a department in the Department filter removes its employees from both charts at the same time, so each department can be studied on its own.

Procedure

1. Create the employee data frame with Employee ID, Department, Years of Service, and Performance Score.
2. Use `plot()` with `type = "b"` to draw the performance line chart and `legend()` to add a legend.
3. Count employees per department with `table()` and use `barplot()` for the bar chart.
4. Use `plot()` for the scatter plot and `abline()` with `lm()` to add a trend line.
5. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts.
6. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.

Result

The employee performance data was successfully visualized using a line chart, bar chart, and scatter plot in R, and an interactive R Shiny dashboard was built combining the line chart and bar chart.

Key R Functions Used

Function	Purpose
<code>data.frame()</code>	Creates the employee dataset.
<code>plot()</code>	Creates the line chart and scatter plot.
<code>legend()</code>	Adds a legend to the line chart.
<code>table()</code>	Counts employees in each department.
<code>barplot()</code>	Creates the department bar chart.
<code>abline()</code> / <code>lm()</code>	Adds a linear trend line to the scatter plot.
<code>fluidPage()</code> / <code>checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws the charts.
<code>shinyApp()</code>	Runs the interactive dashboard.